SP-FSOD: Semantic–Visual Purification-Guided Few-Shot Object Detection on Remote Sensing Images

Jingzheng Tu, Yan Yan, Ying Lu, Cailian Chen, *Senior Member, IEEE*, and Xinping Guan, *Fellow, IEEE*

***Abstract*—Few-shot object detection (FSOD) on remote sensing images is a promising technology for remote sensing applications, as it enhances generalization capability. Most works are devoted to excavating discriminative and compact visual feature representations for FSOD, without considering other domain knowledge. With the rapid development of large language models, promoting FSOD performance with the help of large language models becomes possible. However, the domain gap between text and image is unignorable. In this article, we propose a semantic–visual purification-guided FSOD method for remote sensing images (SP-FSOD) to improve performance. First, a feature purification memory (FPM) is designed to store old knowledge. Second, a semantic–visual triplet loss is proposed to alleviate the domain discrepancy between text and image. Third, a flexible fusion module (FFM) is presented to fuse the feature maps of the semantic and visual domains adaptively. Experiments on large-scale public datasets demonstrate that the proposed SP-FSOD method achieves the state-of-the-art performance on three-, five-, and ten-shot settings.**

***Index Terms*—Few-shot object detection (FSOD), object detection, remote sensing images, semantic–visual correlation.**

I. Introduction

REMOTE object detection serves as a crucial technology in the area of optical remote sensing image (RSI) processing. It aims to accurately locate and classify targets in the input RSIs, which has been applied widely in urban surveillance, transportation planning, land surveying, and real-world applications. Early research was devoted to constructing handcrafted features for remote object detection. With the rapid development of deep learning, researchers adopt convolutional neural network (CNN)-based remote object detection methods to build more compact visual representations. However, these CNN-based remote object detection methods require a large amount of annotated training data, suffering from significant economic and timing costs. Meanwhile, recent research has discovered that most CNN-based methods are not able to function well in realistic applications due to the catastrophic forgetting (a phenomenon that CNNs tend to rapidly and drastically forget previously learned information when learning new information).

Few-shot object detection (FSOD), defined as object detection with a small portion of annotated samples, has emerged as a promising paradigm for RSI processing. It assumes that only k-shot annotated samples are available for novel classes' object detection. FSOD works can be divided into the following main categories: transfer-learning-based, meta-learning-based, and metric-learning-based methods.

Transfer-learning-based methods [1], [2], [3], [4], [5], [6], also called fine-tuning methods, transfer the old knowledge learned from base classes to novel classes via fine-tuning techniques, such as parameter freezing. Note that the network architecture remains the same as generic remote object detection when dealing with base classes and novel classes. In such a manner, the generalization performance on novel classes is improved with only a few samples. However, the domain shift and inconsistency problems are not considered by transfer-learning-based methods.

Meta-learning-based methods [7], [8], [9], [10], [11], [12], [13] concentrate on the few-shot episode construction and prototype extraction from base samples, instead of feature representation building in transfer-learning-based methods. It benefits from fast adaptation to novel classes, but typically requires complex episode training data preparation and training procedures.

Metric-learning-based methods [14], [15], [16] focus on learning similarity metrics between images for object detection. Moreover, building an effective feature representation is challenging due to the cross-image correlation of limited information. Nevertheless, the model bias in base classes and the generalization degradation in novel classes are difficult to trade off [17], [18]. The unbalanced distribution of base and novel classes leads the learned object detection model to

Received 24 August 2025; revised 6 December 2025, 28 February 2026, and 9 May 2026; accepted 7 June 2026. Date of publication 12 June 2026; date of current version 2 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62403443; in part by the Key Laboratory of System Control and Information Processing, Ministry of Education, Shanghai, under Grant Scip20240115; and in part by the National Natural Science Foundation of China under Grant U25B2080, Grant U24B20139, and Grant U244120652. *(Corresponding author: Ying Lu.)*

Jingzheng Tu is with China Academy of Aerospace System and Innovation, Beijing 100088, China, and also with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: tujingzheng@sjtu.edu.cn).

Yan Yan and Ying Lu are with China Academy of Aerospace System and Innovation, Beijing 100088, China (e-mail: 20030626@nuc.edu.cn; jinyuqi0309@sjtu.edu.cn).

Cailian Chen is with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, Shanghai Jiao Tong University, Shanghai 200240, China, also with the School of Automation and Intelligent Sensing, Shanghai Jiao Tong University, Shanghai 200240, China, also with the Shanghai Key Laboratory of Perception and Control in Industrial Network Systems, Shanghai 200240, China, and also with SJTU-Paris Elite Institute of Technology, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: cailianchen@sjtu.edu.cn).

Xinping Guan is with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, Shanghai Jiao Tong University, Shanghai 200240, China, also with the School of Automation and Intelligent Sensing, Shanghai Jiao Tong University, Shanghai 200240, China, and also with the Shanghai Key Laboratory of Perception and Control in Industrial Network Systems, Shanghai 200240, China (e-mail: xpguan@sjtu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3703121

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

prefer the feature distribution of abundant base classes over that of only a few novel classes. In short, most works are dedicated to building elegant feature representations of RSIs and balancing the distinct properties between base classes and the generalization capability on novel classes. The visual domain is attracting nearly the whole attention.

Recently, GPT series large language models provide blasting text generation capability. A question arises naturally: Could large language models help promote visual object detection performance for remote sensing images?

Therefore, in this article, we propose a semantic–visual purification-guided FSOD on remote sensing images, named SP-FSOD. Drawing on the intrinsic correlation between semantic information (e.g., text descriptions of RSI objects) and visual features (e.g., spatial morphology and structure of RSI objects), the proposed SP-FSOD framework is designed to address the limitations above by prioritizing the refinement of both semantic and visual feature representations. Specifically, it first leverages high-quality semantic signals from pretrained large language models to establish a guidance mechanism for visual feature purification, thereby filtering out redundant or noisy visual components irrelevant to the target objects. Concurrently, it optimizes semantic representations by aligning them with discriminative visual cues, ensuring that the semantic information accurately reflects the unique characteristics of RSI objects, such as the elongated shape of runways or the scattered distribution of buildings. Through this bidirectional purification and alignment of semantic and visual features, the framework effectively enhances the discriminability of feature representations for both base and novel classes in RSIs. Ultimately, this integrated design enables SP-FSOD to achieve more effective FSOD in complex RSI scenarios, overcoming the bottlenecks of traditional FSOD methods that rely solely on visual information or fail to fully exploit cross-modal correlations. By leveraging semantic–visual correlation, visual object detection is improved through purified semantic and visual feature representations. The contributions of this article are summarized as follows.

1) A few-shot remote sensing object detection framework based on feature purification memory (FPM) is proposed to boost the learned model's generalization on novel classes.
2) A semantic–visual angular triplet loss is defined to guide model training by a priori descriptions of visual appearance, leveraging on the complementarity of text and image domains.
3) A flexible fusion module (FFM) is designed to integrate the feature representations from semantic and visual domains adaptively.

The remainder of this article is organized as follows. Section II briefly reviews the literature on object detection for computer vision (CV) and remote sensing images. Section IV introduces the proposed method and elucidates each component concretely. The performance of the proposed method is evaluated in Section V. Finally, Section VI concludes this article.

II. Related Works

In this section, the literature review on object detection is given. Specifically, we provide a brief summary of object detection methods in CV and those applied to remote sensing images.

A. Object Detection in CV

Object detection aims to locate and identify objects in a given image, which is a primary research direction in CV. Traditional object detection methods primarily encompass approaches based on feature engineering, sliding windows, and region proposals. Concretely, feature engineering-based methods manually design and extract image features for classifiers or regressors. These methods hinge on professional knowledge and experience and entail substantial manual labor. Sliding window-based methods involve sliding a fixed-size window across an image and using a classifier to determine whether an object exists within the window. This approach suffers from high computational complexity and struggles to handle objects of varying scales. Besides, region proposal-based methods generate candidate object regions and subsequently perform classification and localization on these candidate regions. While these methods have made significant progress in terms of accuracy and efficiency, they still face issues with the inaccurate generation of candidate regions and increased computational complexity.

With the rapid development of deep learning, advanced object detection methods mostly leverage CNNs to extract deep features from raw image pixels. According to the change process of proposals, these deep learning-based methods are mainly categorized into two classes: two-stage detectors and one-stage detectors.

Two-stage detectors typically consist of two stages: region proposal generation and object detection within the generated region proposals. Selective searching [19] is a typical work, in which a bunch of region proposals is generated and then the most negative positions are filtered out in the first stage, and then each region is classified in the second stage. R-CNN [20] substitutes the traditional handcrafted feature extraction process with CNN-based features for performance improvements. Fast R-CNN [21] first generates region proposals and then uses region-of-interest (RoI) pooling before classifying objects for each proposal. After that, Faster R-CNN [22] designs a region proposal network (RPN) to reduce the computational costs of region proposal generation. The RPN network and detection network share weights. Following the idea of Faster R-CNN [22], Mask R-CNN [23] utilizes the feature pyramid network as the backbone for generating multiscale feature maps and introduces mask prediction to detect each instance. However, two-stage detectors typically require more computing resources and time because they involve executing two independent stages.

Moreover, one-stage detectors discard the region proposal generation stage and complete object detection directly on input images. One of the most popular one-stage detectors is YOLO [24], which utilizes a CNN to predict bounding boxes and class probabilities for each object simultaneously.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

Then, a series of improved YOLO are presented, i.e., YOLOv2 [25],$\cdots$, YOLOv10 [26]. Moreover, several SSDs [27], [28], [29] perform object detection using multiscale feature maps. RetinaNet [30] proposes a novel focal loss by reducing the weight of easy negative samples in training. FCOS [31] avoids adjusting hyperparameters and calculations related to anchor boxes. ATSS [32] adaptively selects positive samples to enhance performance. One-stage detectors can maintain a relatively fast speed. However, since predictions are made directly on the image, one-stage detectors are sensitive to changes in the position and scale of targets, resulting in a certain degree of inaccurate localization.

B. Object Detection on Remote Sensing Images

Object detection on remote sensing images and natural scene images both suffer from illumination changes, extremely similar objects, background clutter, etc. Moreover, compared to objects in natural scenes, objects in remote sensing images face additional challenges, including distinct orientations, dynamic scales, extremely small objects, significant intraclass differences, and densely packed arrangements.

Thus, various research makes an effort to address the distinct orientation issue via building rotated bounding boxes [33], [34], [35], [36], intersecting lines [37], and key points [38], [39]. Furthermore, advanced feature embedding [40], [41], [42] and novel loss functions [43], [44] are also proposed to promote object detection performance on remote sensing images.

As far as the dynamic scale issue is concerned, EFPN [45] designs an extended feature pyramid network to use large-scale super-resolution features with rich region information to decouple small-scale object detection. GSDet [46] first incorporates the ground sample distance (GSD) into the object detection modeling process. A GSD identification subnet extracts GSR regression results, which are combined with RoIs to determine object sizes. SFSANet [47] takes the model's scale adaptability into consideration. A scale adaptability module is proposed to strengthen the feature information with the help of the SIoU loss. EMFI-Net [48] and ERPNet [49] consider imperfect saliency maps with coarse boundary details, thus designing an edge-aware multiscale feature integration network and an edge-guided recurrent positioning network for salient object detection in optical RSIs.

Against extremely small objects, MENet [50] proposes a multistage enhancement network to enhance the instance- and feature-level of tiny objects. Besides, AGMF-Net [51] exhibits an attention-free global multiscale fusion network to solve the complex backgrounds and small object detection. Moreover, a teacher–student paradigm-based remote-sensing object detection method [52] is proposed to deal with densely arranged instances. In such a case, a prospective consistency teacher provides supervision information.

Significant intraclass difference is also considered by researchers. DualDA-Net [53] alleviates the domain shift by a dual-head rectification domain adaptation network to exploit unlabeled data. ACNet [54] utilizes neighbor correlation via an adjacent complementary network with global constraints. Features of the current layer and the adjacent layer are aggregated with saliency. In addition, PCLDet [55] presents a prototypical contrastive learning method for fine-grained object detection. Specifically, contrastive learning is introduced to capture discriminative features by increasing interclass distance while decreasing intraclass distance.

To address the densely packed arrangements, ClusDet [56] unifies object clustering and detection in one end-to-end network for small, nonuniformly distributed objects in remote sensing images. DMNet [57] estimates density maps and cropping connected regions based on density masks. Then, the object detection is conducted on the generated foreground. In addition, many studies contribute to other challenges, such as excessive feature coupling [58], [59], unbalanced label distribution [60], diverse aspect ratios [61], [62], [63], etc. Transformer-based GeleNet [64] leverages a direction-aware shuffle weighted spatial attention module and a knowledge transfer module to enhance context interactions.

Consequently, most existing works are devoted to designing discriminative feature representations and salient region clues of RSIs. They nearly all pay attention to the visual domain without considering utilizing the amazing text generation capability of large language models. In this article, we attempt to design a semantic–visual purification-guided method for remote object detection to explore the complementarity of text and image domains.

III. Preliminary

Following previous works for FSOD, we divide the training set into two disjoint sets, i.e., a base set D_{base} and a novel set D_{novel}. The base set D_{base} contains abundant samples with annotations for each class $c \in C_{base}$. Meanwhile, the novel set D_{novel} only includes insufficient annotated samples for each class $c \in C_{novel}$. Note that $C_{base} \cap C_{novel} = \emptyset$.

Each sample in the training set is denoted as $(x, y) \in D_{base} \cup D_{novel}$, where x is an input image and y is the annotations of the image x. Since one input image may contain multiple instances, $y = \{(c_i, b_i), i = 1, \cdots, N\}$. Specifically, $c_i \in C_{base} \cup C_{novel}$ is the class annotations, and b_i is the corresponding bounding box locations. The number of annotated samples per class in C_{novel} is defined as shots.

The primary purpose of FSOD is to improve the generalization capability of object detection algorithms. Specifically, it aims to improve the detection accuracy for novel classes while maintaining the accuracy for base classes.

IV. Methodology

This section first gives an introduction to the overall architecture of the proposed method. Then, each component of the proposed method is illustrated in detail.

A. Overall Architecture

As shown in Fig. 1, the proposed method consists of three main streams: the RoI feature extractor for base classes, the semantic projector for base classes, and the RoI feature extractor for novel classes. The first two streams build visual and semantic representations of base classes, respectively. To alleviate domain gaps between image and text domains,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

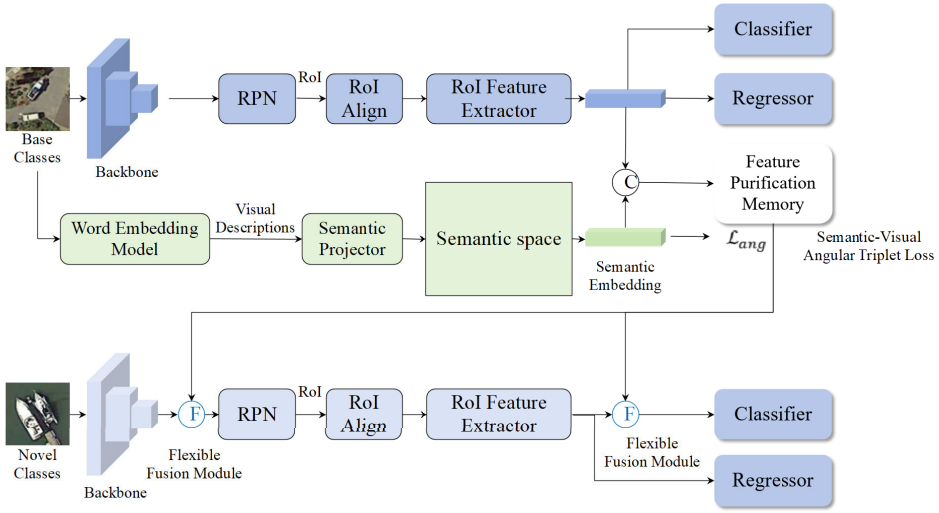


Fig. 1. Overall framework of the proposed SP-FSOD method.

we propose a semantic–visual angular triplet loss (see Section IV-C) that pushes the intraclass distance as small as possible while pulling the interclass distance as large as possible. The outputs of two streams are concatenated to update the FPM (see Section IV-B). Next, the purified feature memory helps the classification and regression of novel classes via an FFM (see Section IV-D).

The workflow of the proposed method is as follows. For base classes, the process starts with extracting visual features via a backbone network. These features then go through an RPN to generate RoI proposals. An RoI align layer refines these proposals, and an RoI feature extractor further processes them. Simultaneously, a word embedding model generates visual descriptions for base-class images, which are projected into a semantic space by a semantic projector. An FPM module, driven by the designed semantic–visual angular triplet loss ($\mathcal{L}_{ang}$), purifies the visual features using semantic embeddings from the semantic space. The purified features are then fed into a classifier for category prediction and a regressor for bounding box regression. For novel classes, an FFM (F) integrates information before the RPN. After RoI align and RoI feature extractor, the FFM processes the features again. Semantic embeddings from the semantic space, shared with the base class pipeline, are used in conjunction with the processed visual features. Finally, a classifier and a regressor perform detection tasks for novel classes, enabling FSOD by leveraging semantic–visual purification and knowledge sharing between base and novel classes.

B. Feature Purification Memory

Storing a small portion of old data preserves the information from old classes while learning new knowledge about novel classes during training, which serves as a promising technique to alleviate catastrophic forgetting. Such an observation raises a question: how to find a proper method for memory management?

With far fewer stored samples than novel classes, memory samples risk overfitting or being overlooked during training.

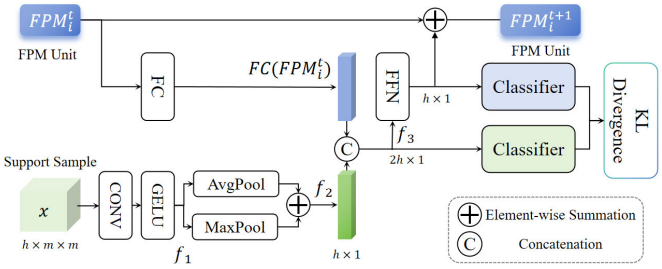


Fig. 2. Schematic of FPM.

Thus, maintaining rich information about the old classes using a small portion of samples is a significant challenge.

Some researchers design a memory mechanism to store the old samples for performance promotion. In this article, we propose an FPM strategy to prevent catastrophic forgetting while learning more effective representations during training. Instead of storing old samples [65], FPM caches representative features of old samples. This design reduces the memory occupancy while preserving key information from old samples.

Fig. 2 illustrates the FPM strategy. "FPM Unit" stores the purified feature memory FPM_i^t at iteration t. The updated FPM_i^{t+1} is used for the next iteration during training to prevent catastrophic forgetting while learning effective representations. Concretely, the support sample x is processed by a convolutional layer and a GeLU operation to build a purified feature representation f_1

$$f_1 = GeLU\left(Conv\left(x\right)\right). \tag{1}$$

Next, we perform average and maximum pooling on f_1 in parallel, then add the features

$$f_2 = AvgPool\left(f_1\right) \oplus MaxPool\left(f_1\right). \tag{2}$$

Then, the summation result f_2 is concatenated with the FPM unit $FPM_i^t \in \mathcal{R}^{m\times 1}$

$$f_3 = Concat\left(f_2, FC\left(FPM_i^t\right)\right). \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

The concatenation f_3 is delivered to a feed-forward network (FFN) to extract the refined feature, which benefits updating the FPM unit via a momentum updating policy [66]

$$FPM_i^{t+1} = \alpha FPM_i^t + (1-\alpha)\, FFN\,(f_3)\,. \tag{4}$$

The FPM unit is regarded as a momentum term at iteration t. $\alpha \in [0,1]$ is the momentum coefficient, controlling the influence of past updates. $(1-\alpha)$ represents the learning rate. To establish more discriminative features, we adopt the KL-divergence and thus

$$\mathcal{L}_{kl} = \mathbb{E}_{f_3}\left[\mathbb{E}_{z\sim\phi(z|f_3)}\left[D_{kl}\left[p\,(y|f_3)\,, p\,(y|FFN\,(f_3))\right]\right]\right]. \tag{5}$$

Specifically, $\mathbb{E}$ is the expectation. ϕ represents the parameters of FFN, and $D_{kl}(\cdot,\cdot)$ is the KL-divergence. y is the label of the support sample. $p(y|\cdot)$ indicates the conditional distribution of the output of the classifier.

The characteristics of FPM include the following.

1) *Enhancing Feature Representation Capability:* FPM extracts rich context information through convolution operations, and purifies features in both channel and spatial dimensions. This can make tiny objects more prominent in conflicting information and help the model learn more discriminative feature representations.
2) *Adaptability to Novel-Class Remote Sensing Images:* Facing the significant target size difference and background clutter, FPM leverages discriminative purified feature representations from old samples, which better adapt to novel-class images.
3) *Alleviating Catastrophic Forgetting:* In few-shot scenarios, limited samples may contain much noise, which can easily lead to model overfitting and then cause catastrophic forgetting. FPM strategy can filter out this kind of noise by using a purified feature representation of images. It retains both semantic-level and fine-grained compatible information, enabling the model to better retain knowledge from base classes and enhancing its generalization ability, thereby reducing the probability of catastrophic forgetting.

C. Semantic–Visual Angular Triplet Loss

An interesting observation is that objects in natural scene images appear visually distinct, yet they can be grouped according to semantic clustering. Naturally, a question arises: Can text-modal descriptions guide and benefit visual object detection?

A straightforward approach is to convert the text descriptions of images into word embedding vectors using encoding strategies. However, this paradigm would suffer from the representation gap problem due to the intrinsic property distinctions between the image and text domains. Thus, in this article, we attempt to embed the semantic–visual similarity into the learning procedure via angular triplet loss.

1) Text Description Generation: When a sample of novel class comes without annotations, the text descriptions are easily obtained via semantic word embeddings. For example, popular large language models (BERT [67], GPT, and Gemini) or typical Word2vec [68] can generate fluent descriptions given an image I_i.

Then, the generated text descriptions $LLM(I_i)$ of input images I_i are embedded by the word embedding model $\mathcal{W}$

$$w_i = \mathcal{W}\,(LLM\,(I_i)) \tag{6}$$

where w_i denotes the word embedding vectors, and $\mathbb{W}$ indicates a chosen word embedding model.

Considering the huge gap between visual and semantic spaces, we treat the common intermediate space as the embedding space to optimize the data structures of both visual features and semantic descriptions. Thus, an alignment function [69], [70], [71] is adopted to alleviate the gap between image and text domains

$$w_i' = \mathbb{G}\,(w_i, v_i)\,. \tag{7}$$

In specific, w_i' is the aligned representation, and v_i denotes the visual features by the RoI feature extractor of image I_i. $\mathbb{G}$ is the alignment function to concatenate semantic features w_i and visual features v_i. The advantages of alignment include: 1) leveraging the information of the text domain and 2) avoiding the inconsistency in text and image embedding spaces.

2) Semantic–Visual Similarity: Text descriptions provide rich and compact information about input images, i.e., shape, color, and textures. Directly encoding them into a semantic embedding brings limited performance improvements. This is because: 1) the domain gap between text and image and 2) targets in remote sensing images have a much smaller size than common objects, thus making it difficult to categorize them against background clutter.

The key to the integration of language models' knowledge with visual features lies in the fact that the embedded semantic vector of one image should try to be consistent with every visual instance feature from this image. Motivated by this, we propose exploring semantic–visual collaboration by building a semantic–visual similarity loss function that leverages semantic information to optimize the visual data structure to be more discriminative in the joint embedding space.

First, a cosine similarity matrix $\hat{S}$ is computed with a self-excluding Softmax operation to establish more discriminative interclass discrepancies

$$S\,(i,j) = \frac{w_i' w_j'}{\|w_i'\|_2 \|w_j'\|_2} \tag{8}$$

$$\hat{S}\,(i,j) = \begin{cases} \frac{exp(S(i,j)/\tau)}{\sum_{j'\neq i} exp(S(i,j')/\tau)}, & if \quad i \neq j \\ S\,(i,j)\,, & otherwise. \end{cases} \tag{9}$$

Now, the normalized similarity matrix $\hat{S}(i,j)$ represents the visual clues between objects. Its diagonal elements are 1, concerning self-similarity. Other elements are $\in (0,1)$, concerning interclass affinities.

3) Loss Function: To establish more discriminative feature representation, we adopt the angular triplet loss function on w_i' by regarding it as an independent feature sample. To begin, a batch-efficient sampling policy is employed to generate both positive and negative samples. Next, the angular triplet loss is conducted on these samples

$$\mathcal{L}_{ang} = \max\left\{0, d\left(w_i', w_{p_i}'\right) - d\left(w_i', w_{n_i}'\right) + \Delta\right\}. \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

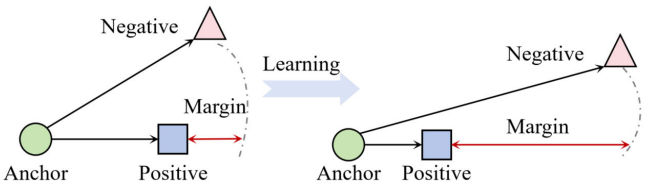


Fig. 3. Schematic of angular triplet loss.

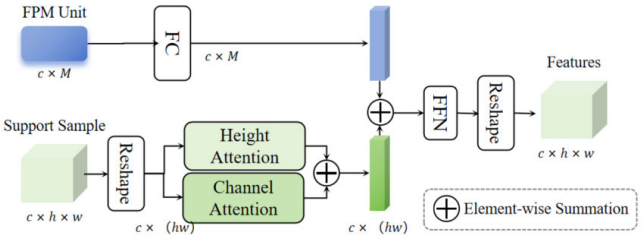


Fig. 4. Architecture of F in Fig. 1, emFFM.

Specifically, $d(\cdot,\cdot)$ is the Euclidean distance. p_i and n_i denote the positive and negative samples for class i. Δ is the margin gap between the minimum interclass distance and the maximum intraclass distance. As shown in Fig. 3, the purpose of angular triplet loss is to push the negative sample as far as possible while pulling the positive samples as near as possible. As a result, each class is more compact in the embedding space.

Thus, we decide to use the visual–semantic similarity to guide the margin of the angular triplet loss

$$\Delta_i = \hat{S}\left(i, p_i\right) - \hat{S}\left(i, n_i\right). \tag{11}$$

Note that Δ is adapted to each class i. Such a design benefits the training procedure by making it steadier and the feature representation more discriminative.

Now, the visual–semantic angular triplet loss is written as follows:

$$\mathcal{L}_{ang}^{i} = \max\left\{0, d\left(w_i', w_{p_i}'\right) - d\left(w_i', w_{n_i}'\right) + \Delta_i\right\} \tag{12}$$

$$\mathcal{L}_{ang} = \sum_{i} \mathcal{L}_{ang}^{i}. \tag{13}$$

The overall loss function is thus the combination of classification loss and the angular triplet loss

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{ang}. \tag{14}$$

In all experiments, we set $\lambda = 0.5$. In the steady training stage, $\mathcal{L}_{ang}$ provides effective cross-modal alignment guidance while ensuring the stability of detection performance.

D. Flexible Fusion Module

To better fuse the memory and the shallow feature of images, we design an FFM to extract salient features with rich regional information. Fig. 4 shows the architecture.

Specifically, the FPM unit provides the old feature from the previous period, denoted as $u_1 \in \mathcal{R}^{c\times M}$. Then, a fully connected layer processes the FPM unit features u_1 and outputs as follows:

$$u_1' = FC\left(u_1\right). \tag{15}$$

Meanwhile, we reshape the features of support sample $u_2 \in \mathcal{R}^{c\times h\times w}$ to $u_2' \in \mathcal{R}^{c\times(hw)}$. Then, u_2' is conducted with height attention and channel attention in parallel

$$u_3 = HAttn\left(u_2'\right) \oplus CAttn\left(u_2'\right) \tag{16}$$

where $\oplus$ denotes elementwise summation. The attention maps are elementwise summed to build a more compact feature map u_3. Then, this feature map u_3 is fused with the memory features u_1'

$$u_f = Reshape\left(FFN\left(u_3 \oplus u_1'\right)\right). \tag{17}$$

The fusion results $u_3 \oplus u_1'$ are sent to an FFN and then reshaped to output the final result u_f.

Compared with summation directly, the advantages of FFM include: *1) richer information:* position attention and channel attention lead the network to focus on the salient clues of feature maps; *2) more flexible fusion:* the features of support samples and memory are fused flexibly instead of decision-level fusion; and *3) less computing complexity:* the size of feature maps after the attention mechanism is smaller, thus the computing complexity is reduced.

In specific, the workflow of how the semantic–visual purification enhances novel-class features is presented as follows.

1) *Step 1:* Knowledge transfer via memory. FPM retains generalized semantic–visual prototypes from base classes. These prototypes act as clean guidance to regularize novel-class features and avoid overfitting to scarce samples.
2) *Step 2:* Noise suppression and purification. The semantic–visual angular triplet loss pulls intraclass features closer and pushes interclass features apart. It removes task-irrelevant visual variation and semantic ambiguity, leaving only stable, category-discriminative components.
3) *Step 3:* Adaptive fusion for novel-class adaptation. FFM adaptively fuses memory-guidance features and novel-class support features. It strengthens shared patterns while preserving novel-category uniqueness, enabling fast adaptation without catastrophic forgetting.

In short, the purified features are compact, generalizable, semantically aligned, and foreground-focused, which directly improves the discrimination and generalization of novel-class remote sensing object detection.

V. Experiments

In this section, extensive experiments are conducted to validate the effectiveness of the proposed method. First, the proposed method is compared with the state-of-the-art methods. Then, an ablation study is conducted to examine the influence of each component. After that, additional experiments were conducted to assess the impact of semantic–visual loss.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NOVEL CLASSES FOR FEW-SHOT TASKS

Datasets	Novel Classes
DOTA-v1.0	airplane, baseball field, tennis court
Fair1M	A350, tugboat, van
DIOR	airplane, baseball field, bridge

A. Datasets and Metrics

DOTA-v1.0 [72] includes 2806 images at 4000 $\times$ 4000 resolution and 188 282 instances. The training set comprises 1411 images, and the validation set contains 458 images. The total images are classified into 16 classes, including plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter, roundabout, soccer ball field, and swimming pool.

Fair1M [73] collects 40 000 images from Gaofen satellites and the Google Earth platform. The scenes depicted in these remote sensing images span from Asia to America and Europe. All images are classified into five categories and 37 subcategories.

DIOR [74] is a large-scale benchmark dataset for optical remote sensing image object detection, covering 20 object classes. DIOR consists of 23 463 remote sensing images and 190 288 object instances. These object instances are manually marked with axis-aligned bounding boxes, and there are 192 472 axis-aligned object bounding box annotations. The image size in the dataset is 800 $\times$ 800 pixels, and the spatial resolution ranges from 0.5 to 30 m. The dataset is divided into a training and validation set (11 725 images) and a test set (11 738 images).

The evaluation metrics are average precision (AP) at an IoU threshold of 0.5, which is abbreviated as AP50 (in %). Table I presents the novel classes of three datasets in the experiment section.

B. Implementation Details

The proposed method is implemented on the PyTorch platform. The hardware platform is an NVIDIA GTX A1000 GPU. The initial learning rate is set as $1e^{-4}$ with a weight decay $2e^{-4}$. The batch size is 32. The text descriptions of images are generated by a Qwen-32B LLM. The text embedding dimension is 300. In all experiments, we set $\lambda = 0.5$ in (14).

C. Performance Evaluation With State of the Art

The performance of the proposed method is evaluated against state-of-the-art methods on large-scale remote sensing image datasets, i.e., *DOTA-v1.0*, **Fair1M**, and **DIOR**.

1) DOTA: Table II presents the comparison results of the proposed method with state-of-the-art methods. Since the comparison methods use different backbones and distinct network architectures, we have reproduced these methods with different backbones for comparison. Compared with RCNN-based methods (Meta RCNN [7] and MM-RCNN [66]), the proposed method achieves 13.3%, 12.8%, and 8.5% higher AP50 in three-, five-, and ten-shot scenarios at least. Furthermore, the proposed method exhibits over 30.0% performance improvement compared to FSCE [3]. Moreover, TEMO [75] obtains 30.9%, 37.2%, and 42.8% AP50 in three-, five-, and ten-shot settings, respectively. ST-FSOD [77] and DPL-Net [76] perform similar, both exceeding the performance of TEMO [75]. Besides, GE-FSOD [78] improves 1.9%, 1.1%, and 3.4% AP50 than ST-FSOD [77] in three settings, respectively. In addition, our method achieves AP50 improvements of 1.1%, 1.0%, and 1.0% over GE-FSOD [78] in three-, five-, and ten-shot scenarios, respectively. The performance comparison results demonstrate the effectiveness of the proposed method. Besides, Fig. 5 illustrates the visualization results of object detection on DOTA.

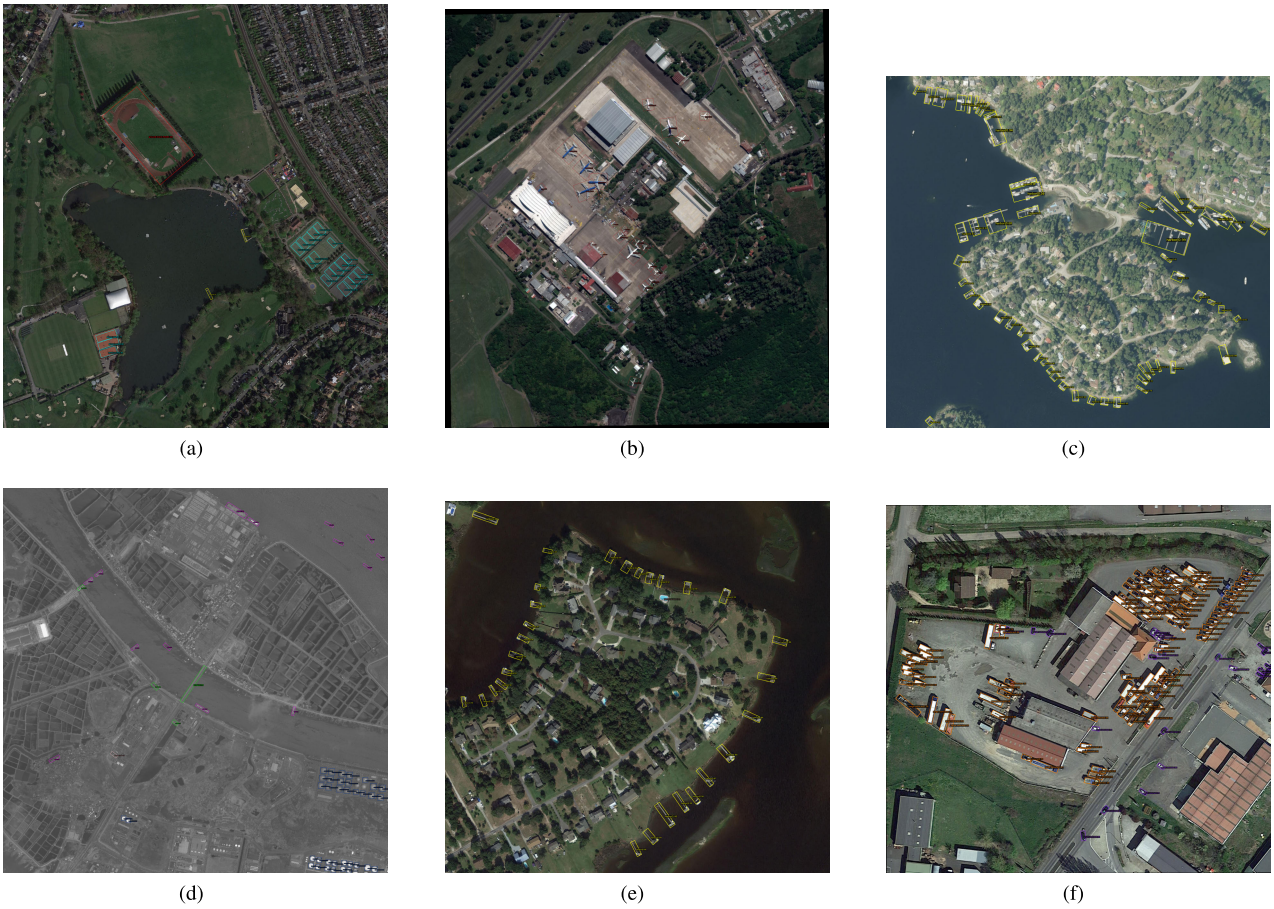
TABLE II
COMPARISON ON DOTA

Shot	Methods	Novel Classes			AP50
		airplane	baseball field	tennis court	
3	Meta RCNN [7]	18.0	61.0	20.0	33.0
	FSCE [3]	17.3	38.7	40.8	24.9
	MM-RCNN [66]	23.0	81.0	24.0	43.0
	TEMO [75]	18.4	47.1	52.4	30.9
	DPL-Net [76]	56.0	71.0	37.0	55.0
	ST-FSOD [77]	55.6	69.5	33.3	53.3
	GE-FSOD [78]	56.5	70.9	37.1	55.2
	SP-FSOD (Ours)	57.2	73.1	38.7	56.3
5	Meta RCNN [7]	50.0	83.0	17.0	50.0
	FSCE [3]	22.8	36.6	49.5	28.3
	MM-RCNN [66]	57.0	89.0	21.0	56.0
	TEMO [75]	24.2	58.9	52.1	37.2
	DPL-Net [76]	69.0	85.0	49.0	68.0
	ST-FSOD [77]	67.4	84.5	50.4	66.7
	GE-FSOD [78]	68.3	83.4	51.6	67.8
	SP-FSOD (Ours)	70.4	85.9	50.1	68.8
10	Meta RCNN [7]	50.0	83.0	45.0	59.0
	FSCE [3]	24.7	56.2	49.1	34.6
	MM-RCNN [66]	63.0	90.0	51.0	68.0
	TEMO [75]	32.6	59.3	67.0	42.8
	DPL-Net [76]	72.0	89.0	62.0	74.0
	ST-FSOD [77]	70.5	84.3	58.6	72.1
	GE-FSOD [78]	70.7	88.8	63.4	75.5
	SP-FSOD (Ours)	73.8	90.0	65.7	76.5

2) Fair1M: Moreover, we conduct a performance comparison between the proposed method and state-of-the-art methods. Table III shows the comparison results. Fig. 6 presents the visualization results of object detection methods on the Fair1M dataset.

The first observation is that frequent occlusions, complex background clutter, and extreme variations in illumination conditions hinder the Fair1M dataset. Nevertheless, the proposed SP-FSOD method performs 28.8%, 29.8%, and 33.4% mAP in three-, five-, and ten-shot scenarios, respectively. This result is 12.1%, 10.6%, and 11.2% higher than two RCNN-based methods, i.e., Meta RCNN [7] and MM-RCNN [66], in three-, five-, and ten-shot scenarios, respectively. Moreover, FSCE [3] and TEMO [75] achieve 22.8% and 27.1% AP50 in a ten-shot setting, which validates the superiority of these two methods. Meanwhile, DPL-Net [76] presents 3.9% higher than TEMO [75] in a ten-shot scenario. Furthermore, ST-FSOD [77] performs better than TEMO [77] with the help of teacher–student self-training. GE-FSOD [78] has a noticeable improvement over ST-FSOD [77] and DPL-Net [76] due to multiscale

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f)

Fig. 5. Visualization object detection results of the proposed method on the DOTA dataset. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

TABLE III
COMPARISON ON FAIR1M

Shot	Methods	Novel Classes			AP50
		A350	Tugboat	Van	
3	Meta RCNN [7]	14.0	8.1	10.0	10.7
	FSCE [3]	21.9	18.0	23.4	21.1
	MM-RCNN [66]	19.8	14.7	15.5	16.7
	TEMO [75]	26.0	23.2	28.6	25.9
	DPL-Net [76]	25.1	21.9	24.0	23.7
	ST-FSOD [77]	24.2	22.0	23.8	23.3
	GE-FSOD [78]	26.5	24.0	25.7	25.4
	SP-FSOD (Ours)	27.1	27.5	31.9	28.8
5	Meta RCNN [7]	16.8	10.2	12.5	13.2
	FSCE [3]	22.3	18.5	25.6	22.1
	MM-RCNN [66]	23.9	16.5	17.2	19.2
	TEMO [75]	26.0	26.3	31.5	27.9
	DPL-Net [76]	29.0	25.8	28.4	27.7
	ST-FSOD [77]	28.4	26.9	27.1	27.5
	GE-FSOD [78]	30.2	26.0	29.2	28.5
	SP-FSOD (Ours)	30.2	29.7	29.5	29.8
10	Meta RCNN [7]	22.0	13.5	14.0	16.5
	FSCE [3]	24.0	19.6	24.8	22.8
	MM-RCNN [66]	28.8	18.2	19.5	22.2
	TEMO [75]	24.7	24.1	32.4	27.1
	DPL-Net [76]	34.0	28.4	30.6	31.0
	ST-FSOD [77]	33.7	27.5	30.1	30.4
	GE-FSOD [78]	34.5	28.5	33.4	32.1
	SP-FSOD (Ours)	34.9	29.8	35.6	33.4

feature representation through dual attention mechanisms and cross-level feature fusion. Moreover, the proposed method achieves 3.4%, 1.3%, and 1.3% improvements in AP50 over GE-FSOD [78] in three-, five-, and ten-shot settings, respectively. This comparison result verifies the effectiveness of the proposed method.

3) DIOR: We compare the proposed SP-FSOD method with other state-of-the-art methods on the DIOR dataset. Table IV presents the experimental results. Meta RCNN [7] adopts a meta-learning framework to enable fast adaptation for FSOD, performing 10.6%, 13.0%, and 14.0% AP50 with three-, five-, and ten-shot settings, respectively. FSCE [3] focuses on feature fusion and class-specific embedding learning. MM-RCNN [66] integrates multimodal information and refines region proposals. TEMO [75] leverages transformer-based architectures for feature encoding. DPL-Net [76] emphasizes dual-path learning for feature representation. Besides, ST-FSOD [77] boosts AP50 by 19.7%, 23.8%, and 29.6% across three settings, leveraging spatial–temporal information. GE-FSOD [78] concentrates on generative methods to augment few-shot samples, which gains 18.1%, 20.5%, and 24.8% AP50 with three settings. Moreover, the proposed SP-FSOD method achieves AP50 of 20.7%, 24.4%, and 30.6% with three-, five-, and ten-shot settings, respectively. This highlights that the semantic–visual purification mechanism in SP-FSOD is highly effective at leveraging additional training samples to refine feature representations, resulting in superior FSOD performance on remote sensing images. Overall,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

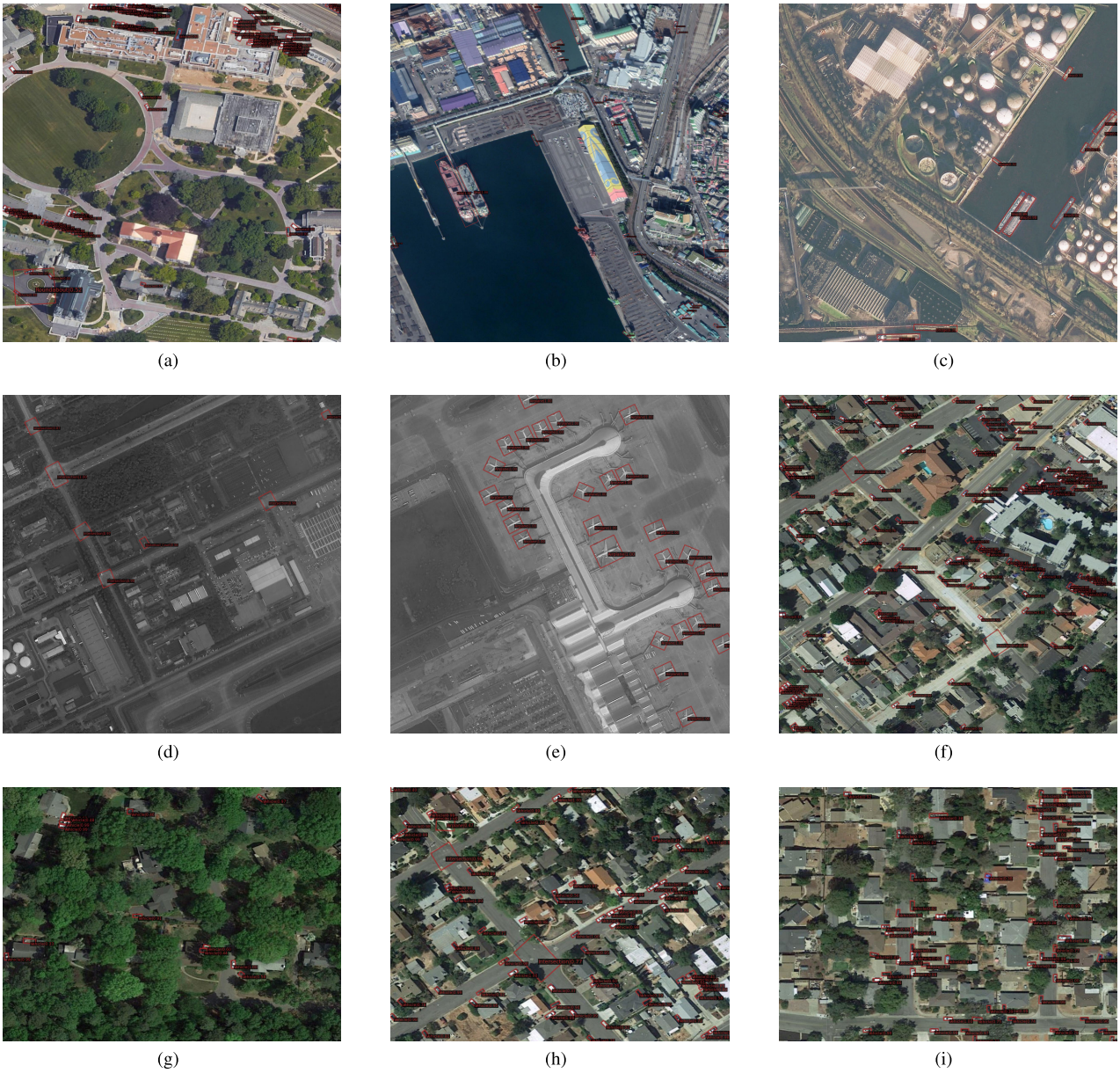


Fig. 6. Visualization of object detection results of the proposed method on the Fair1M dataset. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6. (g) Sample 7. (h) Sample 8. (i) Sample 9.

across all shot settings, SP-FSOD consistently outperforms other methods, validating the effectiveness of the proposed semantic–visual purification-guided approach for FSOD in remote sensing scenarios.

D. Ablation Study

1) Ablation Study of FPM and FFM: Since the gain of FPM and FFM depends on the performance of the semantic–visual angular triplet loss $\mathcal{L}_{ang}$, we first evaluate the effectiveness of FFM and FPM in the proposed SP-FSOD method. Table V presents the AP50 performance in three-, five-, and ten-shot settings on the Fair1M, DOTA, and DIOT datasets, respectively. "Baseline" refers to the proposed method without the FPM and FFM modules and the semantic–visual loss function.

The first observation is that each module contributes to the final performance. Specifically, the FPM module achieves at least 16.1% improvement in AP50 over Fair1M, DOTA, and DIOR in a ten-shot setting. These results validate the FPM module's discriminative feature representation and its retention of old knowledge. Next, the FFM module further boosts AP50 by 3.3%, 9.7%, and 3.1% on average across three datasets. The reason is that FFM fuses old information and the features of support samples using dual attention, resulting in salient feature maps. In short, the ablation study verifies the effectiveness of FPM and FFM in the proposed SP-FSOD method.

2) Ablation Study of Semantic–Visual Angular Triplet Loss: Furthermore, we compare the performance of different losses and examine the effect of the proposed semantic–visual angular triplet loss on the final performance. The network architecture remains the same in this experiment, while only the loss function is changed. Table VI illustrates the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON ON DIOR

Shot	Methods	Novel Classes			AP50
		airplane	baseball field	bridge	
3	Meta RCNN [7]	8.7	9.1	14.0	10.6
	FSCE [3]	13.1	15.6	17.4	15.4
	MM-RCNN [66]	14.1	16.0	16.7	15.6
	TEMO [75]	14.5	16.5	15.2	15.4
	DPL-Net [76]	14.1	20.4	21.3	18.6
	ST-FSOD [77]	17.7	20.9	20.4	19.7
	GE-FSOD [78]	14.9	18.8	20.7	18.1
	SP-FSOD (Ours)	18.1	22.3	21.8	20.7
5	Meta RCNN [7]	10.9	12.3	15.8	13.0
	FSCE [3]	14.0	16.2	20.4	16.9
	MM-RCNN [66]	15.8	23.3	21.0	20.0
	TEMO [75]	14.9	18.8	17.5	17.1
	DPL-Net [76]	16.8	23.1	24.6	21.5
	ST-FSOD [77]	20.7	25.5	25.2	23.8
	GE-FSOD [78]	16.6	22.5	22.4	20.5
	SP-FSOD (Ours)	21.4	26.5	25.3	24.4
10	Meta RCNN [7]	14.9	12.0	15.1	14.0
	FSCE [3]	15.8	23.7	22.2	20.6
	MM-RCNN [66]	20.7	25.2	20.1	22.0
	TEMO [75]	18.9	23.3	18.9	20.4
	DPL-Net [76]	19.6	26.8	29.0	25.1
	ST-FSOD [77]	25.3	31.0	32.4	29.6
	GE-FSOD [78]	19.0	29.4	26.1	24.8
	SP-FSOD (Ours)	27.2	31.5	33.0	30.6

TABLE V
ABLATION STUDY

Shot	Baseline	+ FPM	+ FFM	Fari1M	DOTA	DIOR
3	✓			11.2	13.9	8.4
	✓	✓		25.7	49.6	17.5
	✓	✓	✓	28.8	56.3	20.7
5	✓			13.6	20.8	10.0
	✓	✓		26.3	60.1	21.0
	✓	✓	✓	29.8	68.8	24.4
10	✓			15.0	25.3	12.5
	✓	✓		31.1	62.8	27.9
	✓	✓	✓	33.4	76.5	30.6

TABLE VI
EFFECT OF DIFFERENT LOSSES

Shot	Loss	Fair1M	DOTA	DIOR
3	Cross-entropy Loss	21.2	45.4	14.1
	Least Square Loss [70]	25.3	51.7	17.6
	Region-category Contrastive Loss [71]	27.0	54.2	19.0
	$\mathcal{L}_{ang}$ (Ours)	28.8	56.3	20.7
5	Cross-entropy Loss	22.5	54.5	17.4
	Least Square Loss [70]	26.0	62.0	20.6
	Region-category Contrastive Loss [71]	28.1	65.3	23.5
	$\mathcal{L}_{ang}$ (Ours)	29.8	68.8	24.4
10	Cross-entropy Loss	25.8	58.7	20.4
	Least Square Loss [70]	28.0	64.8	25.9
	Region-category Contrastive Loss [71]	29.5	72.0	27.7
	$\mathcal{L}_{ang}$ (Ours)	33.4	76.5	30.6

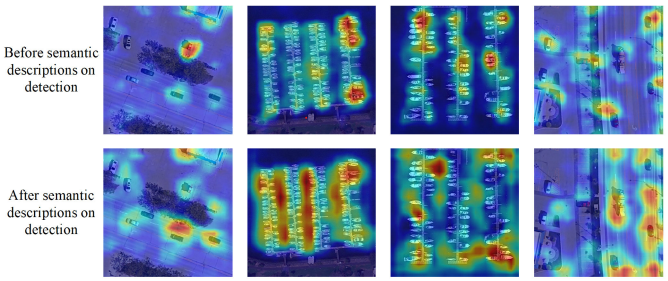


Fig. 7. Visualization results before/after the guidance effect of semantic description on detection.

TABLE VII
EFFICIENCY ANALYSIS ON FAIR1M DATASET

Method	FLOPs	#Params	AP50	FPS
ST-FSOD [77]	/	/	30.4	4.0
GE-FSOD [78]	352.4 GMac	59.9M	32.1	6.5
SP-FSOD w/o FPM	322.6 GMac	54.9M	22.2	7.1
SP-FSOD w/o FFM	382.2 GMac	68.3M	31.1	5.7
SP-FSOD	390.0 GMac	69.5M	33.4	5.6

comparison results on Fair1M, DOTA, and DIOR datasets with three-, five-, and ten-shot settings. The cross-entropy loss only depends on visual information. The listed least square loss [70] and region-category contrastive loss [71] are both semantic–visual cross-modal loss functions. $\mathcal{L}_{ang}$ is the semantic–visual angular triplet loss. The word embedding is the same as the proposed method if the original paper does not specify the implementations.

The previous cross-entropy loss uses only visual information and achieves AP50 scores of 25.8%, 58.7%, and 20.4% on the Fair1M, DOTA, and DIOR datasets in a ten-shot setting. The other three cross-modal loss functions leverage semantic information from image descriptions, and all outperform the cross-entropy loss. This observation validates that semantic information can provide additional performance improvement. Moreover, the least square loss achieves 28.0%, 64.8%, and 25.9% AP50 on three datasets in the ten-shot setting, outperforming the cross-entropy loss. With a more detailed region-category contrastive loss, performance increases by 1.5%, 2.5%, and 1.8% in AP50 across three datasets, respectively. Compared to the previous cross-entropy loss, which only utilizes visual information, the proposed semantic–visual angular triplet loss $\mathcal{L}_{ang}$ achieves 7.5%, 14.3%, and 7.6% AP50 improvements on three datasets on average. This result confirms the superiority of semantic–visual loss and the complementarity between the text and image domains. This observation is helpful for other visual tasks with semantic-domain guidance.

Furthermore, the feature distribution changes before/after the guidance effect of semantic descriptions on detection are shown in Fig. 7. The above four images are feature heatmaps without semantic descriptions for remote object detection, i.e., only visual-domain features. Compared with the above four images without the guidance effect of semantic-domain knowledge, the feature heatmaps of the bottom images with both visual and semantic information focus on more salient subregions of remote objects, including planes, vehicles, and ships. This visualization comparison validates the effectiveness of the semantic descriptions' contribution.

E. Efficiency Analysis

Besides, the efficiency of the proposed SP-FSOD is examined on the Fair1M dataset under a ten-shot setting compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

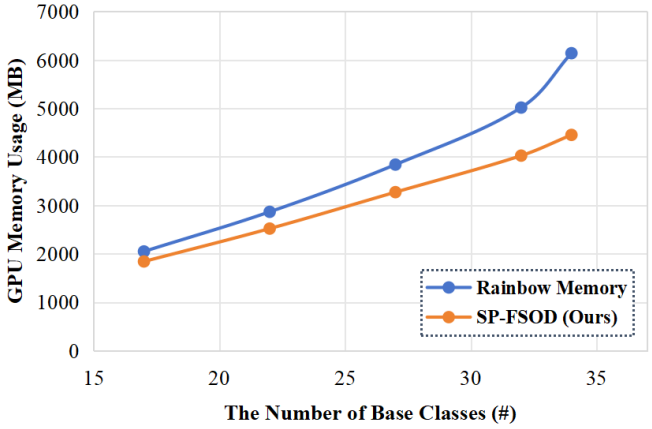


Fig. 8. Effect of different memory strategies on GPU memory usages with increasing numbers of base classes.

with two competitors, i.e., ST-FSOD [77] and GE-FSOD [78]. The results are shown in Table VII. "FLOPs" refers to the number of floating-point operations required for a model to complete one forward propagation, and it is used to measure computational complexity. "#Params" is the total number of parameters, including all weights and bias terms in a model, which directly affects the model size and memory footprint. "FPS" measures the inference speed in frames per second on the test set for each dataset.

First, ST-FSOD [77] achieves 4.0 fps, and GE-FSOD achieves 6.5 fps on the Fair1M dataset. The proposed SP-FSOD method achieves the second-highest FPS on the Fair1M dataset. This validates the competitive efficiency of the proposed method. Moreover, we also examine speed variations within the proposed method's configuration. FPM slows inference speed, while FFM has less impact on efficiency. Moreover, SP-FSOD performs 5.6 fps and 33.4% AP50 on Fair1M. This result confirms that the proposed SP-FSOD method is efficient in real-world applications.

Furthermore, an experiment on the effect of different memory strategies on GPU memory usage as the number of base classes increases is conducted on the Fair1M dataset. We compare the proposed FPM, which stores purified features of old samples, with the rainbow memory strategy [65], which directly stores old samples. Fig. 8 illustrates the results. Both methods use more GPU memory as the number of base classes increases. This result is intuitive because the model needs to maintain a growing volume of data associated with the expanding set of base classes to support incremental learning tasks. Retained information from old classes is used to prevent catastrophic forgetting. This expansion directly translates into higher GPU memory usage, as the GPU must allocate space to store this data during model training and inference. Moreover, another observation is that the GPU memory usage of FPM grows more slowly than that of the rainbow memory strategy. This experimental result validates that feature-level memory storage is a more efficient approach than sample-level memory storage in balancing catastrophic forgetting prevention and GPU memory constraints, supporting the effectiveness of FPM.

TABLE VIII
EFFECT OF DIFFERENT FUSION STRATEGIES

Strategies	Position	Fari1M	DOTA
Gated Fusion	Decision Level	29.3	67.9
Gated Fusion	Feature Level	28.4	66.1
Gated Fusion	Feature + Decision Level	29.9	68.7
Cross-modal Attention	Decision Level	31.2	73.5
Cross-modal Attention	Feature Level	30.6	71.2
Cross-modal Attention	Feature + Decision Level	32.8	75.3
FFM	Decision Level	32.1	73.2
FFM	Feature Level	31.9	70.3
FFM	Feature + Decision Level	33.4	76.5

F. Effect of Different Fusion Strategies

Moreover, a comparison between the proposed FFM and existing adaptive fusion techniques (e.g., gated fusion and cross-modal attention) is conducted on Fair1M and DOTA datasets under a ten-shot setting. Table VIII illustrates the comparison results. "Position" means the placement of the fusion strategies, in which "Decision Level" represents fusion after the RoI feature extractor and "Feature Level" represents fusion before the RPN network. Note that the rest of the network architecture remains the same during this experiment.

With the gated fusion strategy, the highest performance is obtained at 29.9% and 68.7% AP50 on Fair1M and DOTA datasets, respectively. This competitive result validates the fusion capability of gated function. Moreover, using cross-modal attention for fusion achieves AP50 improvements of 2.9% and 6.6% over the gate fusion strategy on two datasets, respectively. Furthermore, FFM introduces a "height attention+channel attention" structure with nondecision-level and decision-level fusions, achieving the best AP50 of 33.4% and 76.5% on Fair1M and DOTA, respectively. The comparison results validate the advantages of FFM.

Besides, we examine the effect of the position selection in fusion strategies. For FFM, feature-level fusion obtains 31.9% and 70.3% AP50 on Fair1M and DOTA, while decision-level fusion achieves 32.1% and 73.2% AP50 on Fair1M and DOTA. This result demonstrates that decision-level fusion is better than feature-level fusion, as it can leverage richer cross-modal information for few-shot remote object detection. Moreover, the combination of feature-level and decision-level fusions achieves the highest AP50 among the three fusion strategies in Table VIII. The conclusion is drawn that the combination of hybrid-level fusion performs better than single-level fusion alone. Thus, we choose the hybrid-level FFM in this article for optimal performance.

G. Discussions

Although the above experiments validate the effectiveness of the proposed SP-FSOD method, it still has several limitations as follows.

1) The reliance on semantic information from predefined word embedding models may restrict the system's adaptability to remote sensing scenarios with highly specialized or novel object categories, as the semantic projector might not fully capture the unique semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

nuances of such cases. Future research could explore more effective ways to control the generation process to achieve more consistent and predictable results.

2) The performance of the FSOD heavily depends on the quality and quantity of training data for both base and novel classes. In situations where few-shot samples are extremely scarce or exhibit severe class imbalance, the model's ability to generalize can be compromised.
3) The computational complexity introduced by the semantic–visual purification process, especially with the FPM and the semantic–visual angular triplet loss, might pose challenges for real-time applications on resource-constrained platforms, limiting its deployment in scenarios that demand low latency.

VI. Conclusion

This article is concerned with FSOD on remote sensing images. A semantic–visual purification-guided method, named SP-FSOD, is proposed to promote generalization to newly coming images. Specifically, an FPM is designed to store partial knowledge of old classes for the recognition of novel classes. Moreover, given the weak correlation between the text and image domains, a semantic–visual angular triplet loss is proposed to leverage semantic–visual interaction information to improve performance. In addition, an FFM is presented to fuse feature maps from text and image domains adaptively. Experiments demonstrate the effectiveness of the proposed method. In the future, we will explore few-shot or zero-shot object detection.

References

[1] F. Zhuang et al., "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jan. 2021.

[2] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 1–8.

[3] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[4] X. W. snd T. Huang, J. Gonzalez, T. Darrell, and F. Yu, "Frustratingly simple few-shot object detection," in *Proc. ICML*, 2020, pp. 9919–9928.

[5] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-shot object detection of remote sensing images via two-stage fine-tuning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[6] Y. Wang, C. Xu, C. Liu, and Z. Li, "Context information refinement for few-shot object detection in remote sensing images," *Remote Sens.*, vol. 14, no. 14, p. 3255, Jul. 2022.

[7] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9576–9585.

[8] B. Kang, Z. Liu, X. Wang, F. Yu, J. Feng, and T. Darrell, "Few-shot object detection via feature reweighting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8419–8428.

[9] L. Karlinsky et al., "RepMet: Representative-based metric learning for classification and few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5197–5206.

[10] T. Liu et al., "Recent few-shot object detection algorithms: A survey with performance comparison," 2022, *arXiv:2203.14205*.

[11] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2022.

[12] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[13] Y. Zhang, B. Zhang, and B. Wang, "Few-shot object detection with self-adaptive global similarity and two-way foreground stimulator in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7263–7276, 2022.

[14] T.-I. Hsieh, Y.-C. Lo, H.-T. Chen, and T.-L. Liu, "One-shot object detection with co-attention and co-excitation," in *Proc. NeurIPS*, 2019, pp. 1–8.

[15] H. Yang et al., "Balanced and hierarchical relation learning for one-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7581–7590.

[16] W. Zhang, C. Dong, J. Zhang, H. Shan, and E. Liu, "Adaptive context- and scale-aware aggregation with feature alignment for one-shot object detection," *Neurocomputing*, vol. 514, pp. 216–230, Dec. 2022.

[17] Z. Fan, Y. Ma, Z. Li, and J. Sun, "Generalized few-shot object detection without forgetting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4527–4536.

[18] C. Lang, G. Cheng, B. Tu, and J. Han, "Learning what not to segment: A new perspective on few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8057–8067.

[19] J. R. R. Uijlings, K. E. A. Van De Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Sep. 2013.

[20] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[21] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[22] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[23] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[24] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[25] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2017, pp. 7263–7271.

[26] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[27] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.

[28] C.-Y. Fu, W. Liu, A. Ranga, A. Tyagi, and A. C. Berg, "DSSD: Deconvolutional single shot detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2017, pp. 1–11.

[29] L. Cui et al., "MDSSD: Multi-scale deconvolutional single shot detector for small objects," *Sci. China Inf. Sci.*, vol. 63, no. 2, pp. 98–100, Feb. 2020.

[30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[31] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[32] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[33] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[34] Y. Jiang et al., "R2CNN: Rotational region CNN for arbitrarily-oriented scene text detection," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 3610–3615.

[35] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[36] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. ICCV*, 2021, pp. 3520–3529.

[37] H. Wei, Y. Zhang, B. Wang, Y. Yang, H. Li, and H. Wang, "X-LineNet: Detecting aircraft in remote sensing images by a pair of intersecting line segments," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1645–1659, Feb. 2021.

[38] P. Dai, S. Yao, Z. Li, S. Zhang, and X. Cao, "ACE: Anchor-free corner evolution for real-time arbitrarily-oriented object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 4076–4089, 2022.

[39] S. He, H. Zou, Y. Wang, B. Li, X. Cao, and N. Jing, "Learning remote sensing object detection with single point supervision," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.

[40] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.
[41] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.
[42] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 4, pp. 3163–3171.
[43] W. Qian, X. Yang, S. Peng, J. Yan, and Y. Guo, "Learning modulated loss for rotated object detection," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 3, pp. 2458–2466.
[44] Z. Zhao and S. Li, "ABFL: Angular boundary discontinuity free loss for arbitrary oriented object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[45] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.
[46] W. Li, W. Wei, and L. Zhang, "GSDet: Object detection in aerial images based on scale reasoning," *IEEE Trans. Image Process.*, vol. 30, pp. 4599–4609, 2021.
[47] Y. Zhang, T. Liu, P. Yu, S. Wang, and R. Tao, "SFSANet: Multiscale object detection in remote sensing image based on semantic fusion and scale adaptability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[48] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.
[49] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.
[50] T. Zhang et al., "Multistage enhancement network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[51] T. Gao, Z. Li, Y. Wen, T. Chen, Q. Niu, and Z. Liu, "Attention-free global multiscale fusion network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[52] J. Shen, C. Zhang, Y. Yuan, and Q. Wang, "Enhancing prospective consistency for semisupervised object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[53] Y. Zhu, X. Sun, W. Diao, H. Wei, and K. Fu, "DualDA-Net: Dual-head rectification for cross-domain object detection of remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612616.
[54] D. Song, Y. Dong, and X. Li, "Adjacent complementary network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[55] L. Ouyang, G. Guo, L. Fang, P. Ghamisi, and J. Yue, "PCLDet: Prototypical contrastive learning for fine-grained object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[56] F. Yang, H. Fan, P. Chu, E. Blasch, and H. Ling, "Clustered object detection in aerial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8311–8320.
[57] C. Li, T. Yang, S. Zhu, C. Chen, and S. Guan, "Density map guided object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 737–746.
[58] K. Fu, Z. Chang, Y. Zhang, and X. Sun, "Point-based estimator for arbitrary-oriented object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4370–4387, May 2021.
[59] E. Liu, Y. Zheng, B. Pan, X. Xu, and Z. Shi, "DCL-Net: Augmenting the capability of classification and localization for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7933–7944, Sep. 2021.
[60] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 526–543.
[61] Y. Zhu, J. Du, and X. Wu, "Adaptive period embedding for representing oriented objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7247–7257, Oct. 2020.
[62] S. Liu, L. Zhang, H. Lu, and Y. He, "Center-boundary dual attention for oriented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.
[63] Y. Zeng, Y. Chen, X. Yang, Q. Li, and J. Yan, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[64] G. Li, Z. Bai, Z. Liu, X. Zhang, and H. Ling, "Salient object detection in optical remote sensing images driven by transformer," *IEEE Trans. Image Process.*, vol. 32, pp. 5257–5269, 2023.
[65] J. Bang, H. Kim, Y. Yoo, J. Ha, and J. Choi, "Rainbow memory: Continual learning with a memory of diverse samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8218–8227.
[66] J. Li et al., "MM-RCNN: Toward few-shot object detection in remote sensing images with meta memory," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[67] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[68] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in *Proc. NeurIPS*, 2013, p. 26.
[69] S. Khandelwal, A. Nambirajan, B. Siddiquie, J. Eledath, and L. Sigal, "Frustratingly simple but effective zero-shot detection and segmentation: Analysis and a strong baseline," 2023, *arXiv:2302.07319*.
[70] L. Zhang, X. Wang, L. Yao, L. Wu, and F. Zheng, "Zero-shot object detection via learning an embedding from semantic space to visual space," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Jul. 2020, pp. 906–912.
[71] C. Yan, X. Chang, M. Luo, H. Liu, X. Zhang, and Q. Zheng, "Semantics-guided contrastive network for zero-shot object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1530–1544, Mar. 2024.
[72] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 3974–3983.
[73] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.
[74] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," 2019, *arXiv:1909.00133*.
[75] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[76] M. Guo, Y. You, and F. Liu, "Discriminative prototype learning for few-shot object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[77] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[78] H. Lin et al., "Generalization-enhanced few-shot object detection in remote sensing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 6, pp. 5445–5460, Jun. 2025.



Jingzheng Tu received the B.Eng. degree from the College of Automation, Xi'an Jiaotong University, Xi'an, China, in 2018, and the Ph.D. degree in information communication and engineering from Shanghai Jiao Tong University, Shanghai, China, in 2023.

She has been with China Academy of Aerospace System and Innovation, Beijing, China, since 2023. Her current research interests include intelligent video analytics of edge-enabled Industrial Internet of Things.



Yan Yan received the B.Eng. degree from the College of Computer Science and Technology, Nanjing University of Science and Technology, Nanjing, China, in 1998, and the Ph.D. degree in control science and engineering from Nanjing University of Science and Technology, in 2012.

She has been with China Academy of Aerospace System and Innovation, Beijing, China, since 2022. Her current research interests include aerospace system integration and large language models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.



Ying Lu received the B.Eng. degree from the College of Automation, Huazhong University of Science and Technology, Wuhan, China, in 2003, and the M.Eng. degree in pattern recognition and intelligent systems from Huazhong University of Science and Technology, in 2006.

He has been with China Academy of Aerospace System and Innovation, Beijing, China, since 2022. He currently serves as the Deputy Director of the Institute of Artificial Intelligence under China Academy of Aerospace System and Innovation and holds the position of the Academic Leader of artificial intelligence with China Aerospace Science and Technology Corporation (CASC), Beijing. His current research interests include cooperative games for aircraft swarm systems and large language models.



Cailian Chen (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees in automatic control from Yanshan University, Qinhuangdao, China, in 2000 and 2002, respectively, and the Ph.D. degree in control and systems from the City University of Hong Kong, SAR, Hong Kong, in 2006.

She has been with the Department of Automation, Shanghai Jiao Tong University, Shanghai, China, since 2008, where she is currently a Distinguished Professor. She has authored three research monographs and more than 100 refereed international journal articles. She is the inventor of more than 30 patents. Her research interests include industrial wireless networks and computational intelligence, and the Internet of Vehicles.

Dr. Chen received the prestigious IEEE TRANSACTIONS ON FUZZY SYSTEMS Outstanding Paper Award in 2008, the IEEE TCCPS Industrial Technical Excellence Award in 2022, and five conference best paper awards. She was awarded the N2Women Star in Computer Networking and Communications in 2022. She won the Second Prize of National Natural Science Award from the State Council of China in 2018, the First Prize of Natural Science Award from the Ministry of Education of China in 2006 and 2016, and the First Prize of Technological Invention of Shanghai Municipal, China, in 2017. She was honored "National Outstanding Young Researcher" by NSF of China in 2020, "Changjiang Young Scholar" in 2015, and China Young Women Scientists Award in 2023. She has been actively involved in various professional services. She is a Distinguished Lecturer of IEEE VTS. She serves as the Deputy Editor for the National Science Open, and an Associate Editor for IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY and *IET Cyber-Physical Systems: Theory and Applications.*



Xinping Guan (Fellow, IEEE) received the B.Sc. degree in mathematics from Harbin Normal University, Harbin, China, in 1986, and the Ph.D. degree in control and systems from Harbin Institute of Technology, Harbin, in 1999.

He is currently a Chair Professor with Shanghai Jiao Tong University, Shanghai, China, where he is the Dean of the School of Electronic, Information and Electrical Engineering, and the Director of the Key Laboratory of Systems Control and Information Processing, Ministry of Education of China. As a Principal Investigator, he has finished/been working on more than 20 national key projects. He is the Leader of the prestigious Innovative Research Team of the National Natural Science Foundation of China (NSFC). His research interests cover industrial network systems, smart manufacturing, and underwater networks.

Dr. Guan is an Executive Committee Member of Chinese Automation Association Council and Chinese Artificial Intelligence Association Council. He was a recipient of the Second Prize of the National Natural Science Award of China in 2008 and 2018, the First Prize of the Natural Science Award from the Ministry of Education of China in both 2006 and 2016, the IEEE TRANSACTIONS ON FUZZY SYSTEMS Outstanding Paper Award in 2008, and the IEEE TCCPS Industrial Technical Excellence Award in 2022. He was honored "National Outstanding Youth" by the NSF of China, "Changjiang Scholar" by the Ministry of Education of China, and "State-Level Scholar" of "New Century Bai Qianwan Talent Program" of China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:33:25 UTC from IEEE Xplore. Restrictions apply.